Equilibrium Concentration of Schottky and Frenkel Defects

Original statement · Problems T2 defects.pdf

> *2. Considering that the energies for Schottky and Frenkel defects formation in a specific material are* $1\ \mathrm{eV}$ *and* $4\ \mathrm{eV}$ *per atom, respectively. Determine the equilibrium concentrations of such defects at* $1000\ \mathrm{K}$*.* $k = 8.62 \times 10^{-5}\ \mathrm{eV/K}$*. (Solution:* 3.026×10^{-3} *and* 8.386×10^{-11}*)*

Solution: 3.026×10^{-3} **and** 8.386×10^{-11}

PHASE 1: HYPOTHESIS & PARAMETERS

Input Data:

- Schottky defect formation energy: $\Delta H_s = 1.0\ \mathrm{eV/defect}$.
- Frenkel defect formation energy: $\Delta H_F = 4.0\ \mathrm{eV/defect}$.
- Absolute temperature: $T = 1000\ \mathrm{K}$.
- Boltzmann constant: $k_B = 8.62 \times 10^{-5}\ \mathrm{eV/K}$.

Starting Hypotheses:

1. The material is a stoichiometric ionic solid of type MX.
2. Defect concentrations are expressed as fractions of the total number of regular lattice sites N (assuming $N_i \approx N$ for the fractional Frenkel concentration $n_F/\quad NN_i \approx n_F/N$).

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

Citing the official formulations for ionic crystals [Session 4 Slide 17]:

1. **Schottky Defects (Anion-Cation Vacancy Pairs):**

The formation of a Schottky pair requires the simultaneous creation of two vacancies of opposite sign. Thermodynamically, the equilibrium fraction is:

$$\frac{n_s}{N} = \exp\left(-\frac{\Delta H_s}{2\,k_B T}\right)$$

2. **Frenkel Defects (Vacancy-Interstitial Pairs):**

The migration of an ion to an interstitial position generates a vacancy and an interstitial. The equilibrium fractional concentration is:

$$\frac{n_F}{N N_i} = \exp\left(-\frac{\Delta H_F}{2\,k_B T}\right)$$

NOTE

The factor 2 in the denominator of the exponent is a fundamental thermodynamic feature of ionic defects: since two defect species are generated simultaneously to preserve electroneutrality, the mass-action product leads to $\exp(-\Delta H / 2 k_B T)$.

PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP

1. Concentration of Schottky Defects (n_s/N):

- Thermal denominator:

$$2\,k_B T = 2 \times (8.62 \times 10^{-5}\ \text{eV/K}) \times (1000\ \text{K}) = 2 \times 0.0862\ \text{eV} = 0.1724\ \text{eV}$$

- Argument of the exponential:

$$-\frac{\Delta H_s}{2\,k_B T} = -\frac{1.0\ \text{eV}}{0.1724\ \text{eV}} = -5.800464$$

- Equilibrium concentration:

$$\frac{n_s}{N} = \exp(-5.800464) = \mathbf{3.0262 \times 10^{-3}} \approx \mathbf{3.026 \times 10^{-3}}$$

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2. Concentration of Frenkel Defects ($n_F/\quad NN_i$):

- Thermal denominator:

$$2\,k_B T = 0.1724\ \text{eV}$$

- Argument of the exponential:

$$-\frac{\Delta H_F}{2\,k_B T} = -\frac{4.0\ \text{eV}}{0.1724\ \text{eV}} = -23.201856$$

- Equilibrium concentration:

$$\frac{n_F}{NN_i} = \exp(-23.201856) = \mathbf{8.386 \times 10^{-11}}$$

PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION

1. **Absolute Thermodynamic Predominance:**

Comparing both populations at 1000 K:

$$\frac{n_s}{n_F} = \frac{3.026 \times 10^{-3}}{8.386 \times 10^{-11}} \approx \mathbf{3.6 \times 10^{7}}$$

The concentration of Schottky defects is more than **36 million times higher** than that of Frenkel defects.

2. **Correlation with Slide 17 [Session 4 Slide 17]:**

As the official theory states verbatim:

$$\text{"In a crystal } \Delta H_s \neq \Delta H_F \implies \text{the defect with the lowest } \Delta H \text{ will form"}$$

Since $\Delta H_s = 1$ eV $\ll \Delta H_F = 4$ eV, the crystal will overwhelmingly present Schottky-type defects, while Frenkel defects will be statistically non-existent.
